BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 009 · Secondary Storage and Device Choice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does RAM hold while a program is running?

Q2. What does a sample rate of 8,000 Hz mean?
A. 8,000 samples per second
B. 8,000 bits per sample
C. 8,000 seconds per sample
D. 8,000 bytes per second

Q3. How does doubling duration affect raw sound size at fixed rate and depth?

Q4. A mono clip uses 2,000 samples/s for 3 s at 8 bits/sample. Calculate its raw size in bytes, excluding headers and compression.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

magnetic — Uses magnetic properties to record bits.

optical — Uses laser light to read disc data.

solid-state — Uses memory chips without moving read/write heads.

secondary storage — Non-volatile storage for saved files and programs.

New learning

Non-volatile storage: Secondary storage keeps saved files and programs without power.

Three technologies: Magnetic HDDs use magnetic patterns; optical discs use lasers; solid-state memory uses chips with no moving mechanical parts.

Comparison: Consider capacity, read/write speed, portability, durability, reliability and cost.

Justification: Link a stated feature of a device to the actual user requirement; no technology is always the best choice.

Worked example

A travelling photographer needs to move large files quickly and often drops their bag. Choose a storage device.

1. Identify speed, portability and physical durability as priorities.

2. Recommend a portable SSD.

3. Explain that an SSD usually transfers large files quickly.

4. Explain why its lack of moving mechanical parts reduces risk of knock damage.

5. Mention that SSDs can cost more per GB than HDDs and backups are still needed.

We Do · Practise with guidance

Guided 1

Recommend storage for very large, low-cost school archives.

Worked method: A magnetic HDD because high-capacity storage is usually economical per GB.

Guided 2

Recommend a small portable storage device for homework.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: USB flash drive or portable SSD; solid-state, compact and portable.

Guided 3

Explain why an expensive fast SSD is not automatically the best solution for every use.

Worked method: Some tasks prioritise price or capacity over speed; user needs determine choice.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict how many whole files fit.

```
capacityGB = 64
fileGB = 8
print(capacityGB DIV fileGB)
```

predict

Question 1

Name two technologies used for secondary storage. [2 marks]

Question 2

Explain one advantage and one disadvantage of an SSD over a magnetic HDD for a portable laptop. [4 marks]

Question 3

A school needs 10 TB for older files. Recommend a suitable device and justify using two characteristics. [4 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

An SSD requires a moving read/write head.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The program's active instructions and working data.

Q2. A — 8,000 samples per second

Q3. It doubles size but does not by itself improve sampling quality.

Q4. (a) 2,000 × 3 × 8 = 48,000 bits; divide by 8 = 6,000 bytes.

I Do / We Do pages · Guided practice

Q1. A magnetic HDD because high-capacity storage is usually economical per GB.

Q2. USB flash drive or portable SSD; solid-state, compact and portable.

Q3. Some tasks prioritise price or capacity over speed; user needs determine choice.

You Do pages · Code First

```
capacityGB = 64
fileGB = 8
print(capacityGB DIV fileGB)
```

Expected result: 8

You Do pages · Independent questions

Q1. Any two: magnetic, optical, solid state [1 each].

Q2. No moving parts [1] so better shock resistance [1]; typically higher cost per GB [1] so high capacity may cost more [1].

Q3. Magnetic HDD [1], high capacity [1] and relatively low cost per GB [1], both supporting an archive workload [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: